

Centre No:

Centre Name:

Code: Subject:

Candidate Number: Candidate Name:

Signature: Evelyn

Do not
write in
this
margin

1. (a) 24
30

(b) 1-2-3-4-go

(c) odd
even

(d) a method that returns void doesn't return data or value, it only perform actions, like int / String
a method that returns a value returns a real data.

2. (a)(i) 1-6
(ii) 2-7

(iii) because there is a loop "for (int i = 0; i < 4; i++)"
the random number is printed out four times

(iv) Random rnd = new Random(); // name the random
rnd.nextInt(6) // use it to pick numbers from 1-6

(b)(i) public static int roll(6) {
 rnd.nextInt(6);
 return rnd;
}

(ii) public static void main (String[] args) {
 Random rnd = new Random();
 for (int i = 0; i < 5; i++) {
 System.out.print (roll(6) + " ");
 }
}

(C) (i) $0.05 \leq x < 1.0$,

(i) ratio of points approximates area ratio.
circle/square $\approx \pi/4$, so $\pi \approx 4 \times \text{cnt}/N$

(ii) 4.0 makes it more accurate for calculation

```
3. (a) public static int nextTerm(int n) {
    if (n % 2 == 0) {
        return n / 2;
    }
    else {
        return 3 * n + 1;
    }
}
```

```
(b) public static int steps(int n) {
    int count = 0;
    while (n != 1) {
        n = nextTerm(n);
        count++;
    }
    return count;
}
```

(c) for needs to claim how many turns a loop will run but while needn't

(d)

```
import java.util.Scanner;
public class CS {
    public static int nextTerm(int n) {
        if (n % 2 == 0) {
            return n / 2;
        }
        else {
            return 3 * n + 1;
        }
    }
    public static int steps(int n) {
```

```
int count = 0;
while (n != 1) {
    n = nextTerm(n);
    count++;
}
return count;
}

public static void main(String[] args) {
    Scanner scan = new Scanner(System.in);
    System.out.print(": ");
    int n = scan.nextInt();
    int current = n;
    while (current != 1) {
        System.out.print(current + " ");
        current = nextTerm(current);
    }
    System.out.print(current);
    System.out.println();
    int stepCount = steps(n);
    System.out.println("Steps: " + stepCount);
    scan.close();
}
}
```